

Full Solution Packet

Translated caps break functional John–Lowner duality for every
positive s

Run `fa.banach.001`, lane 1

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Status: candidate full solution, likely valid

Target: the conjecture in Section 9 of arXiv:2008.09543

1 Source and exact conjecture

Ivanov and Tsiutsiurupa define the John s -function $\mathcal{J}_s f$ and the Lowner s -function $\mathcal{L}_s f$ of a proper log-concave function. On PDF page 22 they ask whether, for every finite $s > 0$, there is an f such that

$$(\mathcal{J}_s f)^\circ \neq \mathcal{L}_s(f^\circ), \tag{1}$$

where $\circ$ denotes log-conjugate polarity [1].

9. DUALITY

Recall that the polar set K° of a body $K \subset \mathbb{R}^d$ is defined by

$$K^\circ = \{y \in \mathbb{R}^d : \forall x \in K \ \langle y, x \rangle \leq 1\}.$$

It is known that for a convex body $K \subset \mathbb{R}^d$ with the John ellipsoid J_K centered at the origin, we have

$$(J_K)^\circ = L_{K^\circ},$$

where L_{K° is the Löwner ellipsoid of K° .

One would expect the same properties for John and Löwner functions (see Subsections 6.1 and 1.1). However, as was shown in [LSW19], there is no duality between the John and Löwner 0-functions. That is, assuming the John 0-function of f is represented by an ellipsoid centered at the origin, we might have $\left({}^{(0)}J_f\right)^\circ \neq {}^{(0)}L_{f^\circ}$. By continuity and Theorem 1.3, it follows that there exist functions f such that $\left({}^{(s)}J_f\right)^\circ \neq {}^{(s)}L_{f^\circ}$ for a sufficiently small positive s . We conjecture that for any positive s there is such an example.

Source evidence: Section 9, PDF page 22. The displayed crop contains the complete conjecture and its surrounding duality statement.

2 Result

For $s > 0$, let

$$\psi_s(t) = \frac{s}{2} \left[\sqrt{1 + 4(t/s)^2} - \log \left(\frac{1 + \sqrt{1 + 4(t/s)^2}}{2} \right) - 1 \right] \quad (t \geq 0). \quad (2)$$

This is the source's radial potential for Lowner s -ellipsoidal functions.

Theorem 1 (Translated-cap counterexample). *Fix any finite $s > 0$ and any $a \in (0, 1)$. In dimension one put*

$$f_{s,a}(x) = (1 - (x - a)^2)_+^{s/2}. \quad (3)$$

Then $f_{s,a}$ and $f_{s,a}^\circ$ are proper integrable log-concave functions, and

$$(\mathcal{J}_s f_{s,a})^\circ \neq \mathcal{L}_s(f_{s,a}^\circ).$$

Consequently the conjecture in (1) is true for every $s > 0$.

3 Proof intuition

The cap in (3) is already a member of the class optimized over in the John problem, so it is its own John function. Translating a function by a multiplies its polar by the exponential tilt e^{-ay} . That tilt is the entire obstruction. Every one-dimensional Lowner s -ellipsoidal density has the same exponential decay rate at the two ends of the line. The tilted polar has decay rates $1 + a$ and $1 - a$, which differ when $a \neq 0$; hence it cannot itself be a Lowner s -ellipsoidal density and cannot equal the Lowner optimizer.

4 Proof

Write

$$h_s(x) = (1 - x^2)_+^{s/2}, \quad \ell_s(y) = e^{-\psi_s(|y|)}.$$

The source's centered polar identity (Lemma 6.1) is

$$h_s^\circ = \ell_s. \quad (4)$$

The function $f_{s,a} = h_s(\cdot - a)$ is a John s -ellipsoidal function. It is feasible in the optimization defining $\mathcal{J}_s f_{s,a}$. Any other feasible nonnegative function g satisfies $g \leq f_{s,a}$ and therefore $\int g \leq \int f_{s,a}$. Thus

$$\mathcal{J}_s f_{s,a} = f_{s,a}. \quad (5)$$

(The uniqueness theorem in the source gives the same conclusion.)

If $q = e^{-\varphi}$ and $q_a(x) = q(x - a)$, then

$$(-\log q_a)^*(y) = ay + \varphi^*(y),$$

so log-conjugate polarity obeys

$$q_a^\circ(y) = e^{-ay} q^\circ(y). \quad (6)$$

Combining (4) and (6) yields the explicit polar

$$f_{s,a}^\circ(y) = e^{-ay - \psi_s(|y|)}. \quad (7)$$

We next check that this function is integrable and is not Lowner s -ellipsoidal. Differentiating (2) gives

$$\psi'_s(t) = \frac{2t}{s(1 + \sqrt{1 + 4(t/s)^2})} \rightarrow 1 \quad (t \rightarrow \infty). \quad (8)$$

Since $0 < a < 1$, equation (8) shows that the negative logarithm $ay + \psi_s(|y|)$ grows linearly with positive rate at both ends. Hence $f_{s,a}^\circ$ is integrable. It is plainly positive, continuous, and log-concave, so the source's existence theorem defines $\mathcal{L}_s(f_{s,a}^\circ)$.

Every one-dimensional Lowner s -ellipsoidal function has the form

$$y \mapsto \alpha e^{-\psi_s(b|y-c|)} \quad (\alpha, b > 0, c \in \mathbb{R}). \quad (9)$$

The derivative of the negative logarithm in (9) tends to b as $y \rightarrow +\infty$ and to $-b$ as $y \rightarrow -\infty$. In contrast, by (7)–(8), the corresponding limits for $f_{s,a}^\circ$ are

$$1 + a \quad (y \rightarrow +\infty), \quad a - 1 \quad (y \rightarrow -\infty). \quad (10)$$

If (7) had the form (9), comparison of the two limits would force

$$b = 1 + a \quad \text{and} \quad b = 1 - a,$$

and hence $a = 0$, contrary to the choice of a . Therefore $f_{s,a}^\circ$ is not Lowner s -ellipsoidal. But $\mathcal{L}_s(f_{s,a}^\circ)$ is Lowner s -ellipsoidal by definition, so they cannot be equal. Together with (5), this proves Theorem 1.

5 Verification report

- **Exact target.** The construction works for every finite $s > 0$, which is precisely the quantifier in the source conjecture.
- **Admissibility.** The translated cap is compactly supported, proper, integrable, and log-concave. Its polar is integrable exactly when $|a| < 1$; choosing $0 < a < 1$ supplies a strict margin at both tails.
- **No optimizer computation.** The proof uses only that every Lowner optimizer belongs to the class (9). Its parameters need not be determined.
- **Dimension.** A one-dimensional example is sufficient for the existential conjecture. No hidden high-dimensional hypothesis is used.
- **Limitation.** The argument uses the finite- s tail limit in (8); it does not assert the same construction at the Gaussian endpoint $s = \infty$.

6 Novelty and human review recommendation

The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2008.09543, functional Lowner ellipsoids, John–Lowner duality, and the exact conjecture; no prior result was found. A bounded external search used the exact conjecture and close phrases. It found the source and the later general optimality paper of Ivanov–Naszodi [2], which emphasizes that functional John and Lowner solutions are generally not dual, but does not state this translated-cap family or explicitly resolve the all- s conjecture. Novelty confidence is moderate rather than absolute.

Send for specialist review. The highest-value checks are the convention in the source's definition of the one-dimensional ellipsoidal classes, the translation identity (6), and the tail comparison (10).

References

- [1] G. Ivanov and I. Tsiutsiurupa, *Functional Lowner Ellipsoids*, arXiv:2008.09543.
- [2] G. Ivanov and M. Naszodi, *Functional John and Lowner Conditions for Pairs of Log-Concave Functions*, arXiv:2212.11781.